



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

ARYAN PATIL

25070122051

CSEA2

Subject: Programming with Java

ASSIGNMENT No : 6

TITLE : Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

Outcome Conclusion Exercise

1. Create a **Vehicle program** where an inner class displays vehicle details and anonymous class performs an action.

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 2 J AS5EX1.java 1 J ASS5EX2.java 3
J AS6EX1.java > Vehicle > main(String[] args)
1  class Vehicle {
2
3      String name = "Car";
4      String model = "Toyota";
5
6      class VehicleDetails {
7          void display() {
8              System.out.println("Vehicle Name: " + name);
9              System.out.println("Vehicle Model: " + model);
10         }
11     }
12
13     Run main | Debug main
14     public static void main(String[] args) {
15
16         Vehicle v = new Vehicle();
17
18         VehicleDetails vd = v.new VehicleDetails();
19         vd.display();
20
21         Runnable action = new Runnable() {
22             public void run() {
23                 System.out.println("Vehicle is Running...");
24             }
25         };
26
27         action.run();
28     }
```

OUTCOME-

```
PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Vehicle Name: Car
Vehicle Model: Toyota
Vehicle is Running...
```

2. Develop a **Food Delivery Application** where an inner class handles order details and anonymous classes handle delivery status updates.

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 2 J AS6EX1.java 2 >
J AS6EX1.java > FoodDelivery > main(String[] args)
1 class FoodDelivery {
2
3     String orderId = "FD101";
4     String foodItem = "Pizza";
5
6     class OrderDetails {
7         void display() {
8             System.out.println("Order ID: " + orderId);
9             System.out.println("Food Item: " + foodItem);
10        }
11    }
12
13    Run main | Debug main
14    public static void main(String[] args) {
15
16        FoodDelivery fd = new FoodDelivery();
17
18        OrderDetails od = fd.new OrderDetails();
19        od.display();
20
21        Runnable deliveryStatus = new Runnable() {
22            public void run() {
23                System.out.println("Order Delivered Successfully!");
24            }
25        };
26
27        deliveryStatus.run();
28    }
```

OUTCOME-

```
PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Order ID: FD101
Food Item: Pizza
Order Delivered Successfully!
```

Conclusion

The Java program successfully demonstrated the use of **inner classes** and **anonymous classes**. The inner class showed how related classes can be logically grouped and how they can access the members of the outer class, improving encapsulation and code organization. The anonymous class demonstrated how a class can be created and used without a separate class definition, making the code shorter and more efficient. This practical helped in understanding the importance of inner and anonymous classes in Java, especially for event handling and object-oriented programming.